

Maternal Health Risk Stratification

Interpretable Machine Learning Analysis of an IoT-Collected Dataset from Rural Bangladesh

StudyBuild – Health & Medical Data Science Track – Project 01

Educational risk-stratification project (not a diagnostic tool)

Dataset	UCI Maternal Health Risk (DOI: 10.24432/C5DP5D)
Cleaned sample size	451 unique records (after duplicate & quality filtering)

Disclaimer. This report is produced solely for educational purposes. The models and findings must not be used for clinical decision-making without further validation, prospective studies and regulatory approval.

Q1 – What does the maternal-risk population look like?

The raw file contains **1 014 rows** and 7 columns. Official UCI documentation reports 1 013 instances; the one-row discrepancy is noted. There are **zero missing values**. However, **562 rows (55.4 %)** are exact duplicates. Two records show a physiologically impossible HeartRate of 7 bpm and were removed.

Cleaning decisions

Two cleaned versions were created for transparency. **Version B** (recommended) removes all exact duplicates and the two HeartRate = 7 records, leaving **451 unique observations**. Extreme ages (Age < 15 or Age > 50) were retained but flagged as unusual for pregnancy.

Risk Level	Count (original)	Prop. (original)	Count (cleaned)	Prop. (cleaned)
low risk	406	0.400	233	0.517
mid risk	336	0.331	106	0.235
high risk	272	0.268	112	0.248

After cleaning, low-risk becomes the majority class (~52 %). High-risk remains a minority class (~25 %). Subsequent modelling therefore emphasises **Recall of the High-risk class**, because missing a true high-risk pregnancy is clinically more costly than other error types.

Summary statistics (Version B)

Statistic	Age	SysBP	DiaBP	BS	Temp	HR
mean	29.2	110.5	75.4	8.35	98.7	74.1
median	25.0	120.0	80.0	7.50	98.0	76.0
min / max	10 / 70	70 / 160	49 / 100	6 / 19	98 / 103	60 / 90

Unusual values documented: HeartRate = 7 (removed); Age extremes retained but flagged; very high BS values (up to 19 mmol/L) kept as clinically plausible in the high-risk group.

Q2 – Which measurements differ most across risk groups?

Group-wise means and medians reveal that **Blood Sugar (BS)** and **Systolic Blood Pressure** produce the strongest separation. High-risk patients show a median BS of approximately 11 mmol/L versus ~7 mmol/L in the other two groups. Mean age is also higher in the high-risk group (~34 years versus ~27–28 years).

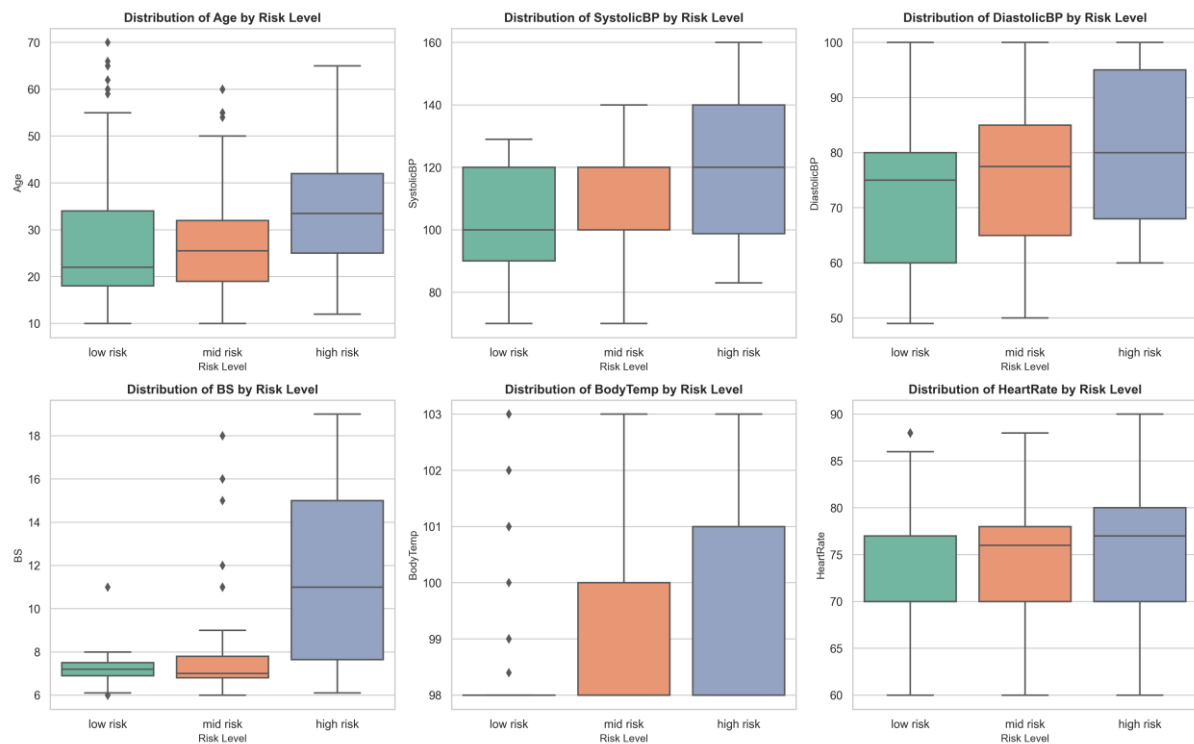


Figure 1. Box-plots of the six clinical variables stratified by RiskLevel (Version B, n = 451). BS and SystolicBP show the clearest group separation; HeartRate and BodyTemp exhibit substantial overlap.

HeartRate and BodyTemp display heavy inter-group overlap and therefore contribute little discriminative information on their own. DiastolicBP shows a mild upward trend but with wide inter-quartile ranges.

Q3 – Which variables appear most strongly associated with High Risk?

A binary indicator IsHighRisk was created and correlated with each continuous feature. Both Pearson and Spearman coefficients rank **BS** first (r_H 0.57 / 0.49), followed by SystolicBP, DiastolicBP, BodyTemp, Age and HeartRate.

Feature	Pearson r	Spearman ρ	Rank
BS	0.573	0.485	1
SystolicBP	0.288	0.273	2
DiastolicBP	0.256	0.231	3
BodyTemp	0.218	0.221	4
Age	0.188	0.216	5
HeartRate	0.182	0.172	6

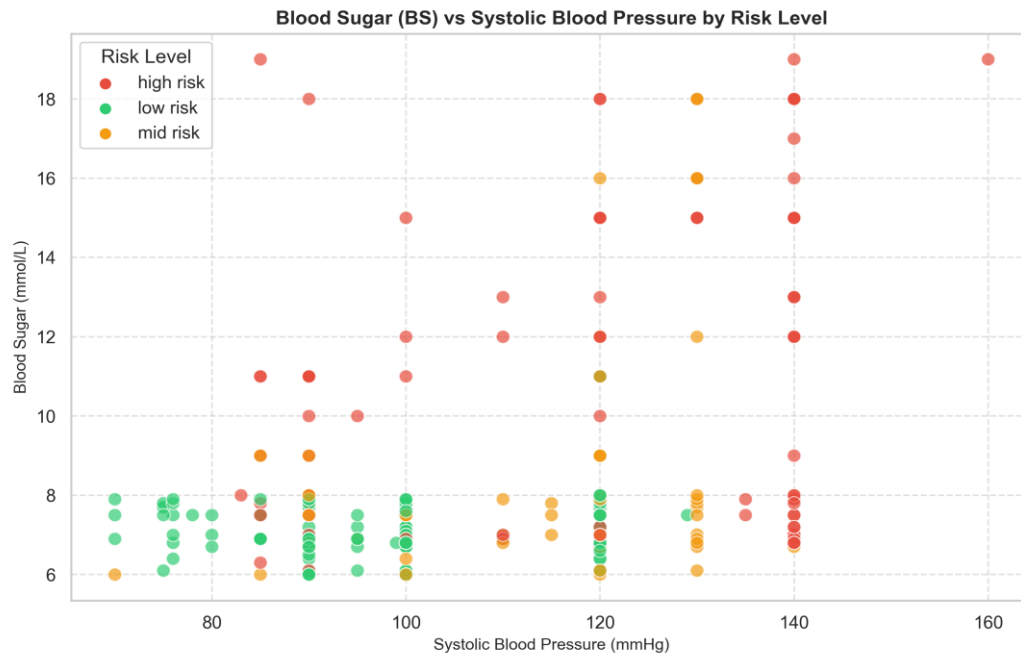


Figure 2. Scatter plot of Blood Sugar versus Systolic Blood Pressure coloured by RiskLevel. High-risk points concentrate in the upper-right quadrant (elevated BS and/or elevated SystolicBP).

Association – causation. The strong correlation of BS with High Risk indicates that elevated blood sugar is a useful *marker* of metabolic disturbance (e.g. gestational diabetes) that raises the probability of adverse outcomes. It does not prove that blood sugar is the sole causal agent.

Q4 & Q5 – Baseline models and why accuracy is not enough

A stratified 80/20 train-test split (random_state = 42) was used. Two simple, interpretable baselines were trained: Logistic Regression (with StandardScaler) and a Decision Tree limited to max_depth = 4. No ensembles or complex hyper-parameter searches were performed, in keeping with project guidelines.

Model / Class	Precision	Recall	F1	Support
Decision Tree (acc = 68.1 %)				
high risk	0.93	0.61	0.74	23
low risk	0.64	0.98	0.77	47
mid risk	0.50	0.10	0.16	21
Logistic Regression (acc = 67.0 %)				
high risk	0.81	0.57	0.67	23
low risk	0.64	1.00	0.78	47
mid risk	1.00	0.05	0.09	21

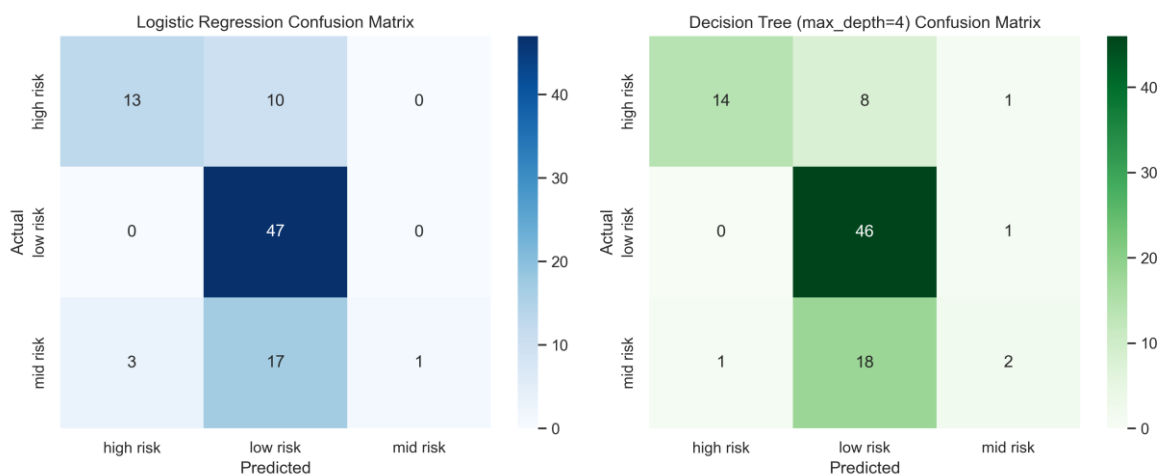


Figure 3. Side-by-side confusion matrices for Logistic Regression (left) and Decision Tree (right).

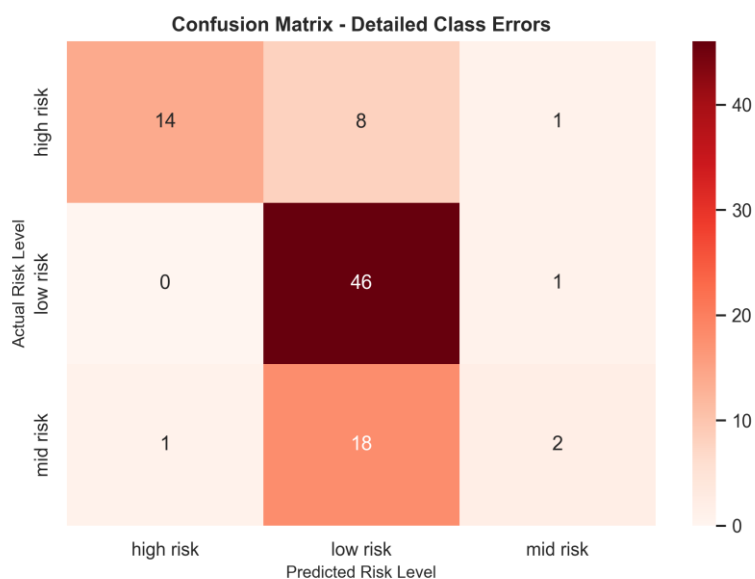


Figure 4. Detailed confusion matrix of the preferred Decision Tree (test set, n = 91).

Why accuracy is insufficient. An overall accuracy of 68 % looks acceptable, yet the model misses approximately 39 % of true High-risk cases (Recall = 0.61) and almost completely fails on the Mid-risk class (Recall H 0.10). In a maternal-health setting a False Negative for High Risk is far more serious than a False Positive; therefore class-wise Recall and F1, especially for High Risk, must be reported and prioritised.

Q6 – Where does the model fail?

Inspection of the 29 misclassified test cases shows a clear pattern dominated by two error pairs:

Actual	Predicted	Count	Share of errors
mid risk	low risk	18	62.1 %
high risk	low risk	8	27.6 %
high risk	mid risk	1	3.5 %
low risk	mid risk	1	3.5 %
mid risk	high risk	1	3.5 %

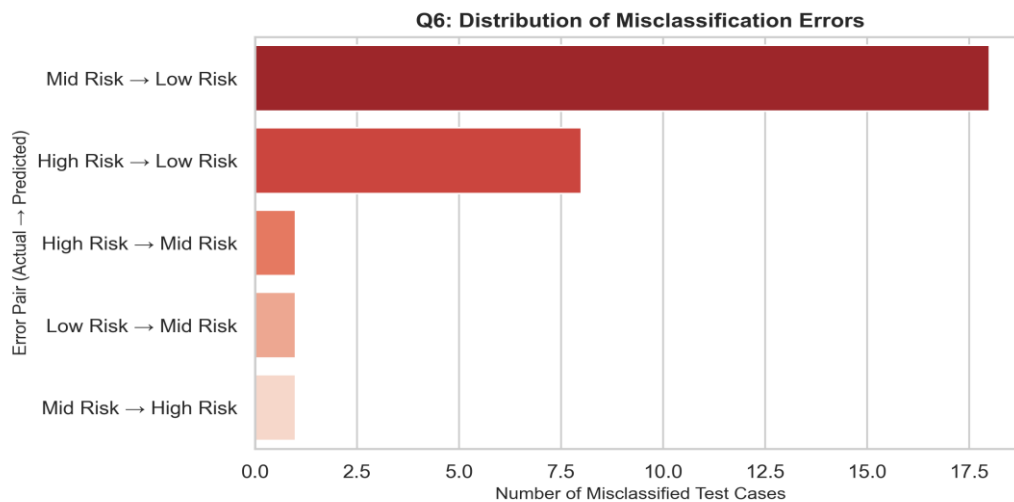


Figure 5. Bar chart of misclassification pairs (Decision Tree, test set). Mid→Low and High→Low dominate the error distribution.

The dominant Mid→Low errors reflect substantial overlap in the measured feature space between these two classes. The clinically more dangerous High→Low errors (8 cases) indicate that a subset of high-risk patients present with biometric profiles that the current feature set cannot reliably distinguish from low-risk profiles.

Q7 – Which features drive the model?

Gini impurity decrease ranks the features used by the Decision Tree as follows:

Feature	Gini Importance	Interpretation
BS	56.8 %	Primary split; metabolic risk marker
SystolicBP	26.4 %	Secondary vascular separator
BodyTemp	10.2 %	Captures febrile presentations
Age	6.7 %	Minor contribution in later splits
DiastolicBP	0.0 %	Unused by the tree
HeartRate	0.0 %	Unused by the tree

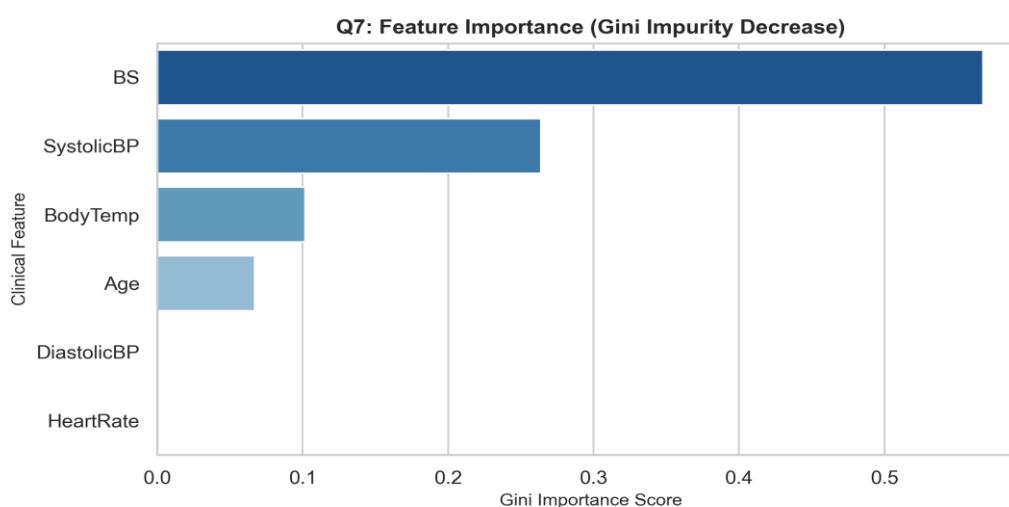


Figure 6. Feature importance (Gini impurity decrease) from the Decision Tree. BS and SystolicBP together account for more than 83 % of the model's splitting power.

These importance scores describe how the particular tree utilises the features; they must not be interpreted as proof of clinical causation. They are nevertheless consistent with known medical associations of hyperglycaemia and hypertension with adverse maternal outcomes.

Q8 – What should a clinician or decision-maker take away?

1. Primary biomarkers. Blood sugar (BS) and systolic blood pressure are the dominant signals associated with higher maternal risk in this dataset. Any screening workflow that ignores these two measurements will lose most of the discriminative information currently available.

2. Model performance & safety. The simple Decision Tree achieves 68 % overall accuracy but only 61 % Recall on the High-risk class. Approximately four out of ten true high-risk cases are labelled low-risk. Consequently the model can serve at best as a *preliminary screening aid* under clinician supervision; it must never replace clinical judgement.

3. Error profile. The majority of mistakes are Mid↔Low confusions caused by overlapping biometric profiles. The second most frequent (and clinically most dangerous) error is High↔Low.

4. Data limitations. The original file contained 55 % exact duplicates; after cleaning only 451 unique records remain. The data originate from a single geographic setting (rural Bangladesh) and lack key clinical covariates such as proteinuria, gestational age, BMI and parity. External validation is absent.

5. Recommendations for future work. Collect the missing clinical variables, enlarge the sample, evaluate cost-sensitive or threshold-tuned models that explicitly optimise High-risk Recall, and conduct prospective external validation before any consideration of real-world deployment.

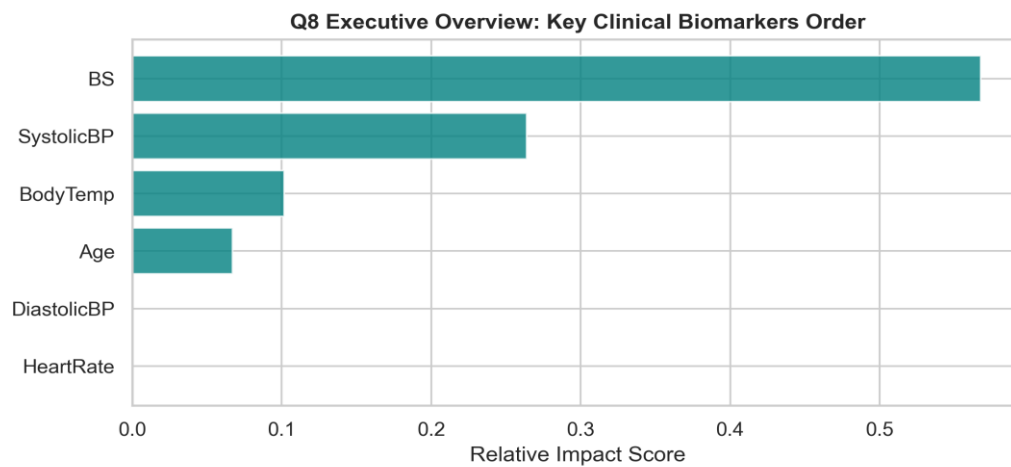


Figure 7. Executive overview of relative clinical biomarker impact derived from the Decision Tree.

End of report. All analyses are fully reproducible from the accompanying Jupyter notebook (*notebooks/analysis.ipynb*) using a fixed random seed (42).